

VisionFood QAI — AI-Powered Quality Inspection for Food & Beverage Manufacturing

Project Review 3: Implementation & Architecture

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Project Phase 1 [AY: 2025–26]

Outline

- 1 Literature Survey
- 2 Technical Details: Uncertainty Quantification and Remediation
- 3 References

Literature Survey — Research Domains

We reviewed **16 peer-reviewed papers** (2024–2025) spanning six research domains in deep learning and industrial quality inspection:

- ❶ CNN-based Defect Detection
- ❷ YOLO Object Detection
- ❸ Food Quality Assessment
- ❹ Industrial Vision Systems
- ❺ IoT & Sensor Fusion
- ❻ Transfer Learning & Anomaly Detection
- ❼ Systematic Review & Framework

Key insight: The literature converges on multi-stage pipelines combining detection + classification, with emerging focus on **explainability**, **incremental learning**, and **supply-chain integration**.

Sources: IEEE ICCAS, ICCSP, GCAT; Elsevier; Vietnam Journal of Computer Science; Foods; J. Imaging

Domain 1 & 2: CNN-Based Defect Detection & YOLO Object Detection

Key Finding: Real-Time Multi-Defect Detection is Feasible

Papers by Chatchapol et al. (ICCAS 2024), Kavitha & Mamatha (IJISRT 2024, GCAT 2024), and Nguyen et al. (VJCS 2025) demonstrate:

- **YOLOv5/v8 achieves >90% accuracy** on real production lines for bottle caps, liquid level, and packaging defects
- **Single-pass detection** (one-stage networks) is preferred over two-stage methods for speed
- **Hardware integration proven:** IoT sensors + PLCs (S7-1200) + embedded systems (Jetson NX) enable end-to-end automation

Highlighted papers:

- Chatchapol et al. — CNN for liquid fill (98.89% accuracy) + conveyor speed control
- Nguyen & Mac — YOLOv8 + multi-threading for real-time classification; scratch defects hardest to detect
- Kavitha et al. — YOLOv5 + Arduino actuation for multi-material sorting (cloth, PVC, PCB)

Domain 3 & 4: Food Quality Assessment & Industrial Vision Systems

Key Finding: Domain-Specific and Comprehensive Multi-Parameter Inspection

Farhangi et al. (CCDS 2024), Sanjay (ICCSP 2024), and Du (MMIA 2025) show:

- **Multi-parameter inspection** (fill level + cap + label + surface) achievable in single pass: **95.6% average accuracy**
- **Food-specific defects** (oil quality, contamination) require specialized algorithms; classical vision + deep learning are complementary
- **Online (real-time) inspection outperforms offline:** 98% accuracy vs. 85–92%, 85% faster processing (3 sec vs. 20 sec per unit)

Highlighted papers:

- Farhangi et al. — LabVIEW + classical vision; 100% cap accuracy, 95% label placement
- Sanjay — CNN for food oil quality; real-time cost-effective alternative to lab testing
- Du — Multi-sensor fusion + edge computing; 60% efficiency gain, 4.85% scrap reduction

Domains 3-7: Food Quality, Industrial Vision, IoT, Transfer Learning, & Systematic Review

Key Findings Across Remaining Domains

- **Food Quality Assessment:** Multi-parameter inspection (fill + cap + label + surface) achievable in single pass with 95.6% accuracy (Farhangi et al.); specialized algorithms for food-specific defects (oil, contamination) (Sanjay)
- **Industrial Vision:** Online real-time inspection outperforms offline: 98% accuracy vs. 85–92%, 85% faster processing (Du); incremental learning enables adaptation to new defect types without full retraining (Ribeiro et al.)
- **IoT & Sensor Fusion:** RFID + LSTM enables item-level spoilage prediction (95.2% accuracy); supply-chain end-to-end monitoring (Cao et al.; Krishnan et al.)
- **Transfer Learning & Anomaly Detection:** Autoencoder-based approaches require only clean samples; knowledge transfer across product types with 3–5 images (Truong & Luong)
- **Systematic Review (Liakos et al., PRISMA 2025):** Fresh produce dominates (36%); beverages only 4% → significant research opportunity. Emerging trends: hyperspectral imaging, explainable AI, blockchain traceability.

Primary Base Paper — Liu et al. (2025)

Research on Mechanical Automatic Food Packaging Defect Detection Based on Improved YOLOv5

Liu et al., PLOS ONE (2025)

Precision: 0.96 | Recall: 0.94 | F1-Score: 0.94

Dataset: 7,400 food packaging images, 5 defect classes

Why Liu et al. is our foundation:

- **Direct relevance:** Multi-defect detection on food packaging using YOLO-based deep learning
- **Proven methodology:** Achieves 94% F1-score on real production data
- **Defect coverage:** 5 classes (scratches, dents, stains, deformations, label errors) — 4 map directly to our taxonomy

Our project extends this work with:

- YOLOv8-based detector for stronger real-time defect detection
- MC Dropout uncertainty quantification for explainable, human-interpretable verdicts

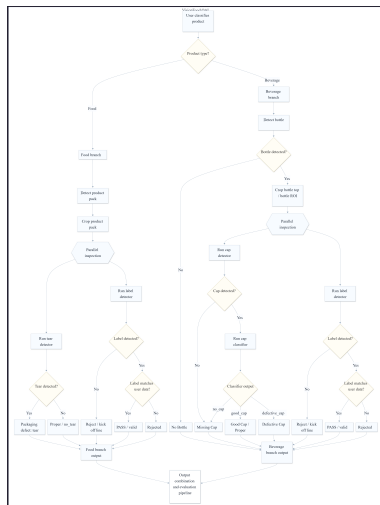
Project Objectives

- Design and deploy a real-time AI-powered quality inspection system for food and beverage manufacturing
- Detect and classify four critical defect types: improper filling, packaging damage, label misalignment, and surface contamination
- Achieve inference latency suitable for production lines (edge-compatible; target: <500 ms per product)
- Integrate with production line hardware (cameras, sensors, PLCs) for automated decision-making

Project Objectives (continued)

- Provide human-interpretable explanations for all automated defect decisions
- Implement uncertainty quantification to escalate low-confidence cases to human review
- Enable incremental learning so the system adapts to new defect patterns without full retraining

Implementation — Food and Beverage Defect Detection Pipeline

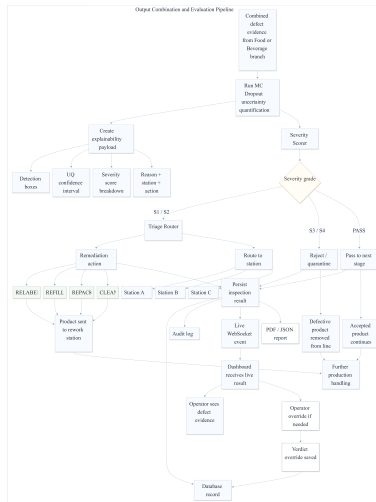


Implementation — Flowchart Terms (Part 1)

Terms used in the first flowchart

- **User classifies product:** the operator selects whether the item is a food product or a beverage.
- **Product type:** the high-level branch that activates the correct pipeline.
- **Detect product pack / Detect bottle:** the first detection stage that finds the product in the image.
- **Crop product pack / Crop bottle top:** isolates the region that needs finer inspection.
- **Parallel inspection:** tear detection and label detection run at the same time.
- **Run tear detector:** checks food packages for packaging damage, mainly tears.
- **Run cap detector:** checks beverage bottles for cap presence.
- **Run cap classifier:** classifies the detected cap as *good_cap*, *defective_cap*, or *no_cap*.
- **Run label detector:** checks whether the label is present and positioned correctly.
- **Label evaluator:** reserved for the user-provided label-matching logic; left blank for now.

Implementation — Output Combination and Evaluation Pipeline



Implementation — Flowchart Terms (Part 2)

Terms used in the second flowchart

- **Combined defect evidence:** the outputs from food or beverage detection are merged into one evaluation stream.
- **MC Dropout uncertainty quantification:** estimates model uncertainty before making a final judgment.
- **Explainability payload:** the evidence shown to the user, including boxes, scores, and reasons.
- **Severity Scorer:** assigns the defect a severity grade based on risk, confidence, area, and retries.
- **Severity grade:** the REMEDY class assigned to the defect, such as S1, S2, S3, or S4.
- **Triage Router:** maps the severity grade to the corrective action.
- **Remediation action:** RELABEL, REFILL, REPACK, or CLEAN for manageable defects.
- **Reject / quarantine:** used for severe or unsafe defects that should not continue on the line.
- **Persist inspection result:** saves the final decision for traceability.
- **Live WebSocket event:** sends the inspection result to the dashboard in real time.
- **Operator override:** allows a human to revise the verdict if needed.

Uncertainty Quantification — MC Dropout Theory

Monte Carlo Dropout estimates model **epistemic uncertainty** (model knowledge gaps) at inference time by keeping dropout layers active:

MC Dropout Methodology

- ➊ **Multiple forward passes:** run the classifier n times on the same input (default: $n = 20$)
- ➋ **Stochastic variation:** each pass applies different random dropout masks
- ➌ **Aggregate outputs:** collect all n predicted confidences for the defect class
- ➍ **Compute statistics:**
 - Mean confidence: $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n c_i$
 - Standard deviation (uncertainty): $\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (c_i - \hat{\mu})^2}$
 - 95% confidence interval: $[\hat{\mu} - 1.96\hat{\sigma}, \hat{\mu} + 1.96\hat{\sigma}]$
- ➎ **Flag uncertain predictions:** if $\hat{\sigma} > \text{threshold (0.15)}$, mark as *uncertain*

Integration with EfficientViT Classifier

- **Model export:** classifier exported to ONNX format *with dropout enabled* (using `training=True` mode)
- **Inference flow:**
 - 1 Extract product crop from detection bounding box
 - 2 Run ONNX session 20 times on the same crop image
 - 3 Collect all 20 confidence scores $\{c_1, c_2, \dots, c_{20}\}$
 - 4 Compute mean, std, and confidence interval from samples
- **Uncertainty thresholds** (tunable via `EdgeConfig`):
 - UQ threshold = 0.15: if $\hat{\sigma} > 0.15$, mark *is_uncertain* = True
 - Escalation threshold = 0.60: if mean confidence < 0.60 *AND* uncertain, escalate to fog tier
- **Result stored in** `UQResult` schema with full statistics for explainability

REMEDY Engine — Architecture Overview

REMEDY (Rapid Evaluation, Mitigation, and Diagnosis of Yield Errors) routes defects to corrective actions:

Input	Component	Output
Detection + UQ	<i>SeverityScorer</i>	Severity Grade (S1–S4)
Grade + Class	<i>TriageRouter</i>	Remediation Action
SKU ID	<i>SKUProfileManager</i>	Config Overrides

Severity Grades

- **S1 (Minor)**: remediable on-line at dedicated station (e.g., relabel, clean)
- **S2 (Moderate)**: requires intermediate handling (e.g., refill, repack)
- **S3 (Severe)**: rejected; does not re-enter production
- **S4 (Critical)**: quarantined; flagged for investigation

REMEDY Severity Scorer — Weighted Formula

Severity score (range [0, 1]) is computed as a weighted sum of four risk components:

$$\text{score} = w_{\text{area}} \cdot a + w_{\text{conf_uq}} \cdot u + w_{\text{class}} \cdot r + w_{\text{attempt}} \cdot p$$

Component Definitions

- a = **Area ratio**: defect bounding box area / total image area (capped at 0.25)
- u = **Uncertainty component**: MC Dropout std dev $\hat{\sigma}$ (capped at 0.30)
- r = **Class risk**: food safety impact of defect class (e.g., contamination=1.0, misalignment=0.30)
- p = **Attempt penalty**: prior remediation failures (capped at 1.0, increases by 0.5 per attempt)

Weights (from EdgeConfig)

$$w_{\text{area}} = 0.35, \quad w_{\text{conf_uq}} = 0.15, \quad w_{\text{class}} = 0.40, \quad w_{\text{attempt}} = 0.10$$

Grade Assignment

Thresholds: S1 if score < 0.30; S2 if < 0.55; S3 if < 0.80; S4 if ≥ 0.80

Triage Routing — Defect-to-Action Mapping

Routing Table: (Defect Class, Severity Grade) → Remediation Action (part 1)

Defect Class	Grade	Action	Station
Label Misalignment	S1/S2	RELABEL	Station A
Improper Filling	S1/S2	REFILL	Station B
Packaging Damage	S1/S2	REPACK	Station C
Surface Contamination	S1	CLEAN	Station C

Triage Routing — Defect-to-Action Mapping (cont.)

Routing Table: (Defect Class, Severity Grade) → Remediation Action (part 2)

Defect Class	Grade	Action	Station
<i>Any class</i>	S3/S4	REJECT	Quarantine
<i>Max attempts exceeded</i>	Any	REJECT	Quarantine
<i>No defects</i>	—	PASS	Shipping

Action Definitions and Station Assignments

- **RELABEL**: re-apply or correct label (Station A)
- **REFILL**: adjust product level or contents (Station B)
- **REPACK**: replace damaged packaging (Station C)
- **CLEAN**: surface decontamination (Station C)
- **REJECT**: remove from line permanently (Quarantine)
- **PASS**: product proceeds to shipping (no intervention)

Each remediation action is routed to its corresponding station, where operators perform the corrective measure. Severe defects (S3/S4) and products exceeding max remediation attempts are automatically rejected, bypassing all stations.

Product Customization — SKU Profiles

SKU Profile Manager loads per-product YAML configurations to override REMEDY parameters:

YAML Configuration Structure

- **class_risk_overrides**: per-class risk weights (e.g., contamination: 0.95, filling: 0.80)
- **rejection_area_thresholds**: maximum defect size before forced rejection per class
- **preferred_stations**: hint for routing (e.g., bottle_250ml prefers Station B for filling checks)
- **uq_threshold_override**: product-specific uncertainty escalation threshold

Example: bottle_250ml.yaml

```
sku_id: "bottle_250ml"
class_risk_overrides:
  improper_filling: 0.85 # Higher risk
  label_misalignment: 0.35
rejection_area_thresholds:
  packaging_damage: 0.10
preferred_stations:
  improper_filling: "B"
```

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Thank You

Questions & Discussion